

Human-preference gain across projection strength

Final checkpoint (1K), sorted from local projection to the full OPSD proposal

Method	Radius	α (token)	α (episode)	Target KL / raw ↓	Pref. margin	Pref. gain	PrefGain / raw ↑	Surplus ↑	Interpretation
Base	0	--	--	0.000	0.061	0.000	0.000	--	Frozen reference
LGSD-Small	0.001	0.339	0.330	0.012	0.141	0.080	1.089	1.078	Minimal target movement
LGSD-Medium	0.004	0.485	0.471	0.053	0.149	0.088	1.194	1.141	Gain grows with radius
LGSD-Large	0.016	0.678	0.662	0.172	0.168	0.107	1.452	1.280	Peak preference gain
LGSD-High	0.040	0.772	0.813	0.389	0.148	0.087	1.190	0.801	Converges toward OPSD
OPSD	Raw	1.000	1.000	1.000	0.134	0.073	1.000	0.000	Full proposal

PrefGain = PrefMargin(model) – PrefMargin(Base). PrefGain/raw divides this gain by matched OPSD. Surplus = PrefGain/raw – Target KL/raw.
 α token is response-token-weighted (paper table); α episode gives every training example equal weight. N=600; no external LLM judge.